

Class #20 - Cramer's Rule

1. Goal: How sensitive is the solution by the changes on the rhs?

To study how the solutions of $Ax = b$ is affected by changes in the entries of b .

2. For any $n \times n$ matrix A and any $b \in \mathbf{R}^n$, let $A_i(b)$ be the matrix obtained from A by replacing column i by the vector b :

$$A_i(b) = \begin{bmatrix} a_1 & \cdots & b & \cdots & a_n \end{bmatrix}$$

Then, $A_i(b) = \begin{bmatrix} Ae_1 & \cdots & \text{_____} & \cdots & Ae_n \end{bmatrix} = A \begin{bmatrix} e_1 & \cdots & \text{_____} & \cdots & e_n \end{bmatrix} = A \cdot \text{_____}$

And, $\det A_i(b) = \det A \cdot \text{_____} = \text{_____}$

3. **Cramer's Rule:** Let A be an invertible $n \times n$ matrix. For any b in $\mathbf{R}^n$, the _____ solution x of $Ax = b$ has entries given by

$$x_i = \text{_____}, \quad i = 1, 2, \dots, n.$$

4. Suppose A is invertible. Then, the j th column of A^{-1} is a vector x that satisfies _____.
By _____,

$$[A^{-1}]_{ij} = \text{_____} = \text{_____}.$$

And, $\det A_i(e_j) = (-1)^{i+j} \det A_{ji} = C_{ji}$.

Thus,

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

The matrix of cofactors is called the *adugate* (or *classical adjoint*) of A , denoted by _____.

Note: The term *adjoint* has another meaning in advanced texts in linear transformations.

Hence,

$$A^{-1} =$$

5. Geometric interpretation of determinants:

Note:

(a) For a 2×2 diagonal matrix:

(b) Let a_1 and a_2 be nonzero vectors in $\mathbf{R}^2$ or $\mathbf{R}^3$. Then for _____ scalar c , the _____ of the parallelogram determined by _____ and _____ equals the area of the parallelogram determined by a_1 and _____.

- (c) Let A be any 2×2 matrix with _____ columns so that the area is *nonzero*.
 A^T can be reduced to a reduced row echelon matrix *without* scaling, i.e., $A^T \sim D$.
 Thus, D can be transformed to A using _____ interchange and _____.
 Hence, area of the parallelogram determined by the columns of A is _____.

Similarly, if A is a 3×3 matrix, the _____ of the _____ determined by the columns of A is _____.

6. Geometric property of linear transformations in $\mathbf{R}^2$ or $\mathbf{R}^3$:

We are interested in how the area (or volume) of $T(S)$ compares with the area (or volume) of the original set S .

Let $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$ be a 2×2 matrix, and S be a region bounded by a parallelogram. Then, S can be written as _____.

Since T is linear, $T(S) =$

The case for 3×3 is analogous.

These hold whenever S is a region in $\mathbf{R}^2$ with finite area or a region in $\mathbf{R}^3$ with finite volume (by dividing the whole region into rectangular subregions).

7. Example: Let S be the tetrahedron in $\mathbf{R}^3$ with vertices at the vectors $0, e_1, e_2$, and e_3 , and let S' be the tetrahedron with vertices at vectors $0, v_1, v_2$, and v_3 . Find a formula for the volume of the tetrahedron S' using a linear transformation that maps S onto S' .